

Assignment 2

Excel Task Report

Profit Percentage:

	A	B	C	D	E	F
1	Date	Sales Rep	Shift	Cost Price	Selling Price	Profit Percentage
2	7/1/2022	Ben	Night	\$1,112.94	\$1,288.00	15.72950923
3	7/1/2022	Jacob	Day	\$1,095.50	\$1,250.75	14.17161114
4	7/4/2022	Ben	Night	\$1,120.94	\$1,295.00	15.52803897
5	7/4/2022	Jacob	Day	\$1,509.73	\$1,687.75	11.79151239
6	7/5/2022	Ben	Day	\$1,235.33	\$1,412.25	14.32167923
7	7/5/2022	Jacob	Night	\$881.93	\$962.00	9.078951844
8	7/6/2022	Ben	Night	\$1,075.70	\$1,142.75	6.233150507
9	7/6/2022	Jacob	Day	\$714.58	\$800.50	12.02384618
10	7/7/2022	Ben	Day	\$1,299.11	\$1,360.00	4.687054984
11	7/7/2022	Jacob	Night	\$1,087.48	\$1,268.50	16.64582337
12	7/8/2022	Ben	Night	\$803.79	\$907.75	12.93372647
13	7/8/2022	Jacob	Day	\$1,113.73	\$1,278.75	14.81687662
14	7/11/2022	Ben	Day	\$843.16	\$965.25	14.48005124
15	7/11/2022	Jacob	Night	\$1,552.52	\$1,717.50	10.62659418
16	7/12/2022	Ben	Night	\$1,382.27	\$1,550.25	12.15247383
17	7/12/2022	Jacob	Day	\$1,515.76	\$1,622.75	7.058505304
18	7/13/2022	Ben	Day	\$1,264.56	\$1,328.50	5.056304169
19	7/13/2022	Jacob	Night	\$1,251.98	\$1,402.00	11.98261953
20	7/14/2022	Ben	Night	\$986.11	\$1,054.00	6.884627476
21	7/14/2022	Jacob	Day	\$1,514.20	\$1,571.25	3.767666094
22	7/15/2022	Ben	Day	\$1,318.85	\$1,454.75	10.30443189
23	7/15/2022	Jacob	Night	\$1,495.11	\$1,655.00	10.69419641
24	7/18/2022	Ben	Night	\$1,187.70	\$1,297.75	9.265807864
25	7/18/2022	Jacob	Day	\$1,150.74	\$1,225.75	6.518414238
26	7/19/2022	Ben	Day	\$1,562.68	\$1,736.75	11.13919676
27	7/19/2022	Jacob	Night	\$1,514.13	\$1,620.25	7.008645229
28	7/20/2022	Ben	Night	\$778.27	\$930.25	19.52792733
29	7/20/2022	Jacob	Day	\$1,270.66	\$1,334.75	5.043835487
30	7/21/2022	Ben	Day	\$766.72	\$841.75	9.785840985
31	7/21/2022	Jacob	Night	\$1,041.13	\$1,157.25	11.15326616
32	7/22/2022	Ben	Night	\$1,218.08	\$1,344.00	10.33758045
33	7/22/2022	Jacob	Day	\$1,016.36	\$1,108.25	9.041087804
34	7/25/2022	Ben	Day	\$1,576.91	\$1,683.00	6.727714327
35	7/25/2022	Jacob	Night	\$870.02	\$997.00	14.59506678
36	7/26/2022	Ben	Night	\$1,145.26	\$1,344.25	17.37509387

	A	B	C	D	E	F
37	7/26/2022	Jacob	Day	\$1,313.05	\$1,504.00	14.54247744
38	7/27/2022	Ben	Day	\$833.49	\$906.50	8.759553204
39	7/27/2022	Jacob	Night	\$1,578.85	\$1,711.75	8.41751908
40	7/28/2022	Ben	Night	\$1,314.78	\$1,371.75	4.333044312
41	7/28/2022	Jacob	Day	\$972.81	\$1,139.75	17.16059662
42	7/29/2022	Ben	Day	\$1,551.16	\$1,721.25	10.96534207
43	7/29/2022	Jacob	Night	\$1,366.59	\$1,418.50	3.79850577
44	8/1/2022	Ben	Night	\$1,466.66	\$1,624.75	10.77891263
45	8/1/2022	Jacob	Day	\$1,594.15	\$1,682.25	5.526456105
46	8/2/2022	Ben	Night	\$1,580.98	\$1,679.00	6.199951929
47	8/2/2022	Jacob	Day	\$916.97	\$1,025.00	11.78119241
48	8/3/2022	Ben	Night	\$726.12	\$838.00	15.40792156
49	8/3/2022	Jacob	Day	\$1,461.44	\$1,599.50	9.446846945
50	8/4/2022	Ben	Day	\$1,176.26	\$1,263.25	7.39547379
51	8/4/2022	Jacob	Night	\$825.12	\$1,004.00	21.67927089
52	8/5/2022	Ben	Night	\$1,034.97	\$1,091.00	5.413683488
53	8/5/2022	Jacob	Day	\$939.94	\$1,045.00	11.17730919
54	8/8/2022	Ben	Day	\$1,429.63	\$1,570.75	9.871085526
55	8/8/2022	Jacob	Night	\$1,009.06	\$1,189.00	17.83243811
56	8/9/2022	Ben	Night	\$974.45	\$1,152.50	18.27184566
57	8/9/2022	Jacob	Day	\$1,511.87	\$1,688.75	11.6994186
58	8/10/2022	Ben	Night	\$1,103.76	\$1,256.75	13.86080307
59	8/10/2022	Jacob	Day	\$809.07	\$930.00	14.94679076
60	8/11/2022	Ben	Night	\$1,438.40	\$1,560.50	8.488598443
61	8/11/2022	Jacob	Day	\$1,472.96	\$1,637.00	11.13675864
62	8/12/2022	Ben	Day	\$1,556.29	\$1,742.25	11.94892983
63	8/12/2022	Jacob	Night	\$960.13	\$1,088.25	13.34402633
64	8/15/2022	Ben	Night	\$1,520.43	\$1,594.50	4.871648152
65	8/15/2022	Jacob	Day	\$1,195.05	\$1,263.00	5.685954563
66	8/16/2022	Ben	Night	\$1,357.83	\$1,518.75	11.85126268
67	8/16/2022	Jacob	Day	\$1,440.38	\$1,498.50	4.035046307
68	8/17/2022	Ben	Night	\$801.34	\$913.25	13.96535803
69	8/17/2022	Jacob	Day	\$1,001.99	\$1,097.00	9.48213056
70	8/18/2022	Ben	Day	\$1,121.86	\$1,191.75	6.229832599
71	8/18/2022	Jacob	Night	\$776.22	\$961.25	23.83731416
72	8/19/2022	Ben	Night	\$779.66	\$900.75	15.53112895

	A	B	C	D	E	F
73	8/19/2022	Jacob	Day	\$850.14	\$932.25	9.658409203
74	8/22/2022	Ben	Day	\$896.25	\$1,008.25	12.49651325
75	8/22/2022	Jacob	Night	\$1,051.00	\$1,133.00	7.802093245
76	8/23/2022	Ben	Night	\$1,385.50	\$1,504.50	8.588957055
77	8/23/2022	Jacob	Day	\$1,075.76	\$1,262.75	17.38212984
78	8/24/2022	Ben	Day	\$896.05	\$957.00	6.802075777
79	8/24/2022	Jacob	Night	\$843.34	\$1,039.25	23.23025114
80	8/25/2022	Ben	Night	\$1,211.66	\$1,325.75	9.416007791
81	8/25/2022	Jacob	Day	\$1,357.20	\$1,463.25	7.813881521
82	8/26/2022	Ben	Night	\$940.87	\$1,027.75	9.234006823
83	8/26/2022	Jacob	Day	\$1,274.05	\$1,455.00	14.2027393
84	8/29/2022	Ben	Night	\$1,179.06	\$1,376.00	16.7031364
85	8/29/2022	Jacob	Day	\$774.20	\$946.25	22.22293981
86	8/30/2022	Ben	Day	\$1,350.48	\$1,509.50	11.77507257
87	8/30/2022	Jacob	Night	\$1,053.37	\$1,136.25	7.868080542
88	8/31/2022	Ben	Night	\$1,161.92	\$1,237.00	6.461718535
89	8/31/2022	Jacob	Day	\$951.93	\$1,136.00	19.33650584

Here, the formula used is:

$=((E_n - D_n) / D_n) * 100$, where $n=\{2,3,...,89\}$

Where, E_n is the selling price, D_n is the cost price and $(E_n - D_n)$ is the profit.

The general formula is:

Profit percentage = (Profit/Cost price) * 100

Number of shifts for each Employee:

Ben:

Day shift:

=COUNTIFS (B2:B89, "Ben", C2:C89, "Day")

Night shift:

=COUNTIFS (B2:B89, "Ben", C2:C89, "Night")

Jacob:

Day shift:

=COUNTIFS (B2:B89, "Jacob", C2:C89, "Day")

Night shift:

=COUNTIFS (B2:B89, "Jacob", C2:C89, "Night")

Where, B2:B89 is the criteria range 1, C2:C89 is the criteria range 2 and inside the quotation marks is the criteria for each range respectively.

The general formula is:

=COUNTIFS (criteria_range1, criteria1, [criteria_range2, criteria2], ...)

Here, first criteria range 1 is checked for criteria(condition) if it is true then criteria range 2 is checked for criteria(condition) if it is also true then count number is increased by one. One by one all rows in range are checked. Then the total count number is displayed as result.

Total Income for Jacob:

=SUMIFS(J2:J89,B2:B89,"Jacob",C2:C89,"Day")											
	I	J	K	L	M	N	O	P	Q	R	S
Rate	Commission Amount	Total Earnings	Ben	Day Shift	Night Shift	Jacob	Day Shift	Night Shift	Jacobs Income	Day	Night
0.50%	\$0.88	\$175.94		17	27		27	17		3381.296	2314.248

Day shift:

Here, the formula used is:

=SUMIFS (J2:J89, B2:B89,"Jacob", C2:C89,"Day")

Where, J2:J89 is the range for column of Jacob's Total Earning of a single item in each row, B2:B89 is the criteria range 1 column and C2:C89 is the criteria range 2 column.

Night shift:

Here, the formula used is:

=SUMIFS (J2:J89, B2:B89,"Jacob", C2:C89,"Night")

Where, J2:J89 is the range for column of Jacob's Total Earning of a single item in each row, B2:B89 is the criteria range 1 column and C2:C89 is the criteria range 2 column.

Here, is first condition in criteria range 1 column is true, which is to check for Jacob's name, then the second condition is checked in criteria range 2 column, which is to check for Day or Night, if it is also true then the value in J2:J89 is added to sum. Then the total sum is displayed at the end.

Commission for Sales:

H2	=IFS(AND(F2>0%,F2<12%),0%,AND(F2>=12%,F2<=20%),0.5%,F2>20%,1.5%)									
	A	B	C	D	E	F	G	H	I	J
1	Date	Sales Rep	Shift	Cost Price	Selling Price	Profit Percentage	Profit	Commission Rate	Commission Amount	Total Earnings
2	7/1/2022	Ben	Night	\$1,112.94	\$1,288.00	15.73%	\$175.06	0.50%	\$0.88	\$175.94
3	7/1/2022	Jacob	Day	\$1,095.50	\$1,250.75	14.17%	\$155.25	0.50%	\$0.78	\$156.03
4	7/4/2022	Ben	Night	\$1,120.94	\$1,295.00	15.53%	\$174.06	0.50%	\$0.87	\$174.93
5	7/4/2022	Jacob	Day	\$1,509.73	\$1,687.75	11.79%	\$178.02	0.00%	\$0.00	\$178.02
6	7/5/2022	Ben	Day	\$1,235.33	\$1,412.25	14.32%	\$176.92	0.50%	\$0.88	\$177.80
7	7/5/2022	Jacob	Night	\$881.93	\$962.00	9.08%	\$80.07	0.00%	\$0.00	\$80.07
8	7/6/2022	Ben	Night	\$1,075.70	\$1,142.75	6.23%	\$67.05	0.00%	\$0.00	\$67.05
9	7/6/2022	Jacob	Day	\$714.58	\$800.50	12.02%	\$85.92	0.50%	\$0.43	\$86.35
10	7/7/2022	Ben	Day	\$1,299.11	\$1,360.00	4.69%	\$60.89	0.00%	\$0.00	\$60.89
11	7/7/2022	Jacob	Night	\$1,087.48	\$1,268.50	16.65%	\$181.02	0.50%	\$0.91	\$181.93
12	7/8/2022	Ben	Night	\$803.79	\$907.75	12.93%	\$103.96	0.50%	\$0.52	\$104.48
13	7/8/2022	Jacob	Day	\$1,113.73	\$1,278.75	14.82%	\$165.02	0.50%	\$0.83	\$165.85
14	7/11/2022	Ben	Day	\$843.16	\$965.25	14.48%	\$122.09	0.50%	\$0.61	\$122.70
15	7/11/2022	Jacob	Night	\$1,552.52	\$1,717.50	10.63%	\$164.98	0.00%	\$0.00	\$164.98
16	7/12/2022	Ben	Night	\$1,382.27	\$1,550.25	12.15%	\$167.98	0.50%	\$0.84	\$168.82
17	7/12/2022	Jacob	Day	\$1,515.76	\$1,622.75	7.06%	\$106.99	0.00%	\$0.00	\$106.99

First Commission rate is found by the following conditions:

- If profit percentage is less than 12% then, commission rate is equal to 0%.
- If profit percentage is greater than or equal to 12% less than or equal to 20% then, commission rate is equal to 0.5%
- If profit percentage is greater than 20% then, commission rate is equal to 1.5%.

Here, formula used is:

=IFS(AND(F2>0%,F2<12%),0%,AND(F2>=12%,F2<=20%),0.5%,F2>20%,1.5%)

Here IFS function is used running multiple conditions, AND is used for more than one logical conditions to check for. The arguments in AND are conditions to be checked and if they are true then the argument next to it is the value which will be assigned to the cell if the conditions are true. In this way, the same process is repeated for other conditions.

12	∨	:	✕	✓	f _x	=G2*H2				
	A	B	C	D	E	F	G	H	I	J
1	Date	Sales Rep	Shift	Cost Price	Selling Price	Profit Percentage	Profit	Commission Rate	Commission Amount	Total Earnings
2	7/1/2022	Ben	Night	\$1,112.94	\$1,288.00	15.73%	\$175.06	0.50%	\$0.88	\$175.94
3	7/1/2022	Jacob	Day	\$1,095.50	\$1,250.75	14.17%	\$155.25	0.50%	\$0.78	\$156.03
4	7/4/2022	Ben	Night	\$1,120.94	\$1,295.00	15.53%	\$174.06	0.50%	\$0.87	\$174.93
5	7/4/2022	Jacob	Day	\$1,509.73	\$1,687.75	11.79%	\$178.02	0.00%	\$0.00	\$178.02
6	7/5/2022	Ben	Day	\$1,235.33	\$1,412.25	14.32%	\$176.92	0.50%	\$0.88	\$177.80
7	7/5/2022	Jacob	Night	\$881.93	\$962.00	9.08%	\$80.07	0.00%	\$0.00	\$80.07
8	7/6/2022	Ben	Night	\$1,075.70	\$1,142.75	6.23%	\$67.05	0.00%	\$0.00	\$67.05
9	7/6/2022	Jacob	Day	\$714.58	\$800.50	12.02%	\$85.92	0.50%	\$0.43	\$86.35
10	7/7/2022	Ben	Day	\$1,299.11	\$1,360.00	4.69%	\$60.89	0.00%	\$0.00	\$60.89
11	7/7/2022	Jacob	Night	\$1,087.48	\$1,268.50	16.65%	\$181.02	0.50%	\$0.91	\$181.93
12	7/8/2022	Ben	Night	\$803.79	\$907.75	12.93%	\$103.96	0.50%	\$0.52	\$104.48
13	7/8/2022	Jacob	Day	\$1,113.73	\$1,278.75	14.82%	\$165.02	0.50%	\$0.83	\$165.85
14	7/11/2022	Ben	Day	\$843.16	\$965.25	14.48%	\$122.09	0.50%	\$0.61	\$122.70
15	7/11/2022	Jacob	Night	\$1,552.52	\$1,717.50	10.63%	\$164.98	0.00%	\$0.00	\$164.98
16	7/12/2022	Ben	Night	\$1,382.27	\$1,550.25	12.15%	\$167.98	0.50%	\$0.84	\$168.82

Now Secondly, the commission amount is calculated by simply multiplying the commission rate with profit.

Here, formula used is:

$$=G_n:H_n \quad , \text{ where, } n=(2, 3, 4, \dots, 89)$$